Experiment 10 — Alumnex Source Study, Python/MySQL Lab and Testing

Description

Experiment 10 — UI, Database, Unit Testing and Integration Testing

Faculty requirement

Develop the user interface using Python, create a MySQL database, and perform unit and integration testing.

Important stack distinction

The submitted Alumnex Connect application itself is **not Python/MySQL**. The inspected repository is a React + Node.js/Express + MongoDB/Mongoose project, uses Redis/Socket.IO for real-time scaling, and has a Capacitor mobile client. The faculty's Python/MySQL requirement has been prepared as a separate educational account-registration/login lab inside the downloadable artifact package; it must not be presented as Alumnex production code.

Actual Alumnex Connect source and architecture

Details

Assignee

THIRUMURUGAN A 24CSR330

Assign to me

Parent

ACS-28 Alumnex Connect — Scrum Lab Ex

Sprint

Add sprint

Labels

experiment-10 scrum-lab

Due date

Add due date

Team

Add team

- Repository: FSD/alumni-connect-website-main at master · SrimanKumarV/FSD
- Live frontend entry point: AlumniConnect - Student-Alumni Platform
- Backend service: https://alumnex-backend-9y5t.onrender.com
- Frontend page/router evidence: App.js
- API wiring and persistence: backend/server.js, User model
- Actual database models include User, Mentorship, Job, JobApplication, DevProfile, ActivityGoal/Record, Message, Notification, Event, Google Calendar connection/event, ForumPost, Project, Contest and related operational models.
- Test sources include authService, RBAC, mentorship service, job service, Activity Hub/reliability, Google Calendar, mobile OAuth and FCM notification tests.
- CI test workflow defines backend Jest tests, frontend production build and Playwright smoke checks. README reports 56 backend tests; that number is a repository statement, not a fresh run from this environment.

Deployment evidence available during this review

Render service metadata reports the frontend and backend services as not suspended and their most recent deploys as live from commit `e1e684b5814fbf9fcb194cae6fcfd6bb4773954e` on 11 October 2026. GitHub's check-runs for that commit show the Android build workflow completed successfully (frontend bundle and APK build steps passed), while a Cloudflare Pages check failed. These are source/deployment signals, not a substitute for manually interacting with every live feature.

The current runtime could not resolve the live demo host for an interactive browser pass. Therefore no claim is made that the production user journey, external OAuth,

Start date

Add date

Story point estimate

Add story points

Reporter

SV SRIMAN VEL

Development

Connect development tools

Open in coding tool

Create branch

Create commit

Automation

Recent rule runs Refresh

Refresh to see recent runs.

Create new automation rule

Created 1 day ago

Updated 31 minutes ago

Configure

email delivery, real-time chat, FCM push or Google Calendar sync was manually completed during this task.

Separate faculty Python/MySQL lab

Package: `ACS_Scrum_Experiments_Artifacts.zip` (provided separately in the conversation).

Implementation in `python_mysql_lab/`:

- Python WSGI UI with registration/login forms and JSON endpoints.
- MySQL schema `schema.sql` for `acs_student_lab.users`.
- Parameterized SQL, normalized email, field validation, duplicate email rejection, PBKDF2-HMAC-SHA256 password hashing and safe error handling.
- Unit tests and offline WSGI/fake-DB integration tests.

Latest offline suite documented on 14 October 2026:
`PYTHONPATH=. python -m unittest discover -s tests -v`

16 tests discovered, 15 passed, 1 skipped, 0 failures. The skipped case is the live-MySQL registration/login integration test. The current execution environment lacks a reachable/configured MySQL server and the connector package, so true MySQL integration remains **NOT VERIFIED**. This is tracked in ACS-35. The package contains `setup_xampp.md` for running that test from a configured Windows/XAMPP lab.

Lab subtask breakdown\n\n* ACS-22 — registration validation and duplicate rejection (lab code).\n* ACS-23 — register endpoint contract (lab code).\n* ACS-24 — unit/integration test plan.\n* ACS-25 — credential verification (lab code).\n* ACS-27 — authentication test

cases.\n* ACS-34 — offline test evidence.\n* ACS-35 — run against real MySQL; remains To Do.\n\n## Acceptance checklist

- [x] Alumnex source architecture reviewed and linked.
- [x] Python UI and MySQL schema prepared as a separate faculty lab.
- [x] Unit and offline request-to-data-access tests run and recorded.
- [ ] Run the Python lab against a reachable real MySQL instance and record successful console output.
- [ ] Perform a live Alumnex browser smoke test with permitted test account(s); record results per feature, not passwords/tokens.
- [ ] Do not mark this experiment Done until the real MySQL test required by the faculty brief succeeds and evidence is captured.

Attachments

Add attachment | Create project plan Try

Subtasks

85% Done

Work		P...	A...	Sta
ACS-22	Registration — Define validation and duplicate checks	=	SV	Do
ACS-23	Registration — Implement and document endpoint contract	=	SV	Do

Work	P...	A...	Sta
ACS-24 Registration — Unit and integration test plan	=	SV	Do
ACS-25 Login — Credential verification	=	SV	Do
ACS-27 Login — Authentication test cases	=	SV	Do
ACS-34 Experiment 10 — Offline Test Evidence	^	SV	Do

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

SV

Add a comment...

Suggest a reply... | Status update... | Thanks...

Pro tip: press M to comment

SV **SRIMAN VEL**
1 hour ago

Verification update — 14 October 2026

Re-ran the offline suite after adding tests for unknown-user authentication failure and database-connection failure.

Command: `PYTHONPATH=. python -m unittest discover -s tests -v`

Result: **16 tests discovered, 15 passed, 1 skipped, 0 failures.**

The skipped test is the real-MySQL registration/login integration test. This runtime has no local MySQL server/client and no `mysql-connector-python` installed, so the live database acceptance criterion remains open. The updated source package includes `setup_xampp.md` for running that test from a Windows/XAMPP lab. Do not treat the offline fake-DB tests as proof of a real MySQL integration.

SV **SRIMAN VEL**
1 day ago

Verification update (13 Oct 2026): the refreshed source package was executed locally. `PYTHONPATH=. python -m unittest discover -s tests -v` reports 14 tests, 13 passed, 1 skipped, 0 failures. The skipped test is the real MySQL integration test; it remains pending until a reachable MySQL instance and driver are configured. Source archive is provided separately in the conversation because the Atlassian Media upload endpoint could not be reached from this runtime.

SV **SRIMAN VEL**
1 day ago

Test checkpoint: five pure validation unit tests passed in the available Python workspace. The two opt-in Flask/MySQL integration cases were skipped; Flask/MySQL could not be installed/configured here. Live API persistence is not verified and must not be represented as passing.

